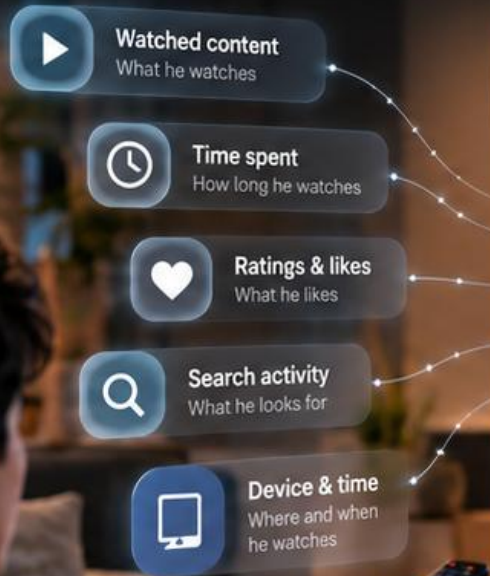


How data creates value



After a long day at work, Victor loves to watch his favorite shows. Media streaming companies use advanced data analytics to **understand his viewing habits and preferences**.

Based on these insights, they can provide him with **personalized recommendations** and **create new shows** that will appeal to him and other customers like him.



See how Victor's actions correlate to one media streaming company's approach to data.

1

Collect data

The platform collects data from Victor's viewing activity, search history, ratings, and device usage.



2

Analyze patterns

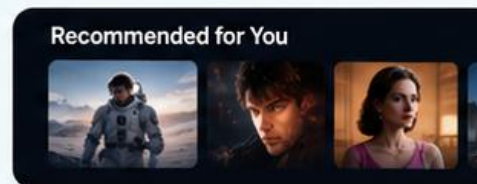
Advanced analytics identify his preferences, viewing habits, and emerging interests.



3

Deliver personalized value

The platform recommends relevant content that matches his interests.



4

Create new content

Insights from millions of viewers like Victor help the company develop new shows and better content.



1

VICTOR'S ACTION

Continuing to watch while working out

One day, while working out, Victor continues watching the most recent episode on his mobile phone.



STREAMING COMPANY'S APPROACH

Requesting data

Victor can seamlessly switch across devices because the company keeps a user profile of his activities across devices and integrates them into one record that is synchronized and stored in the cloud.

2

VICTOR'S ACTION

Searching for new shows

A few weeks later, Victor wants to watch a similar show, so he searches for new content on his tablet.



STREAMING COMPANY'S APPROACH

Analyzing data

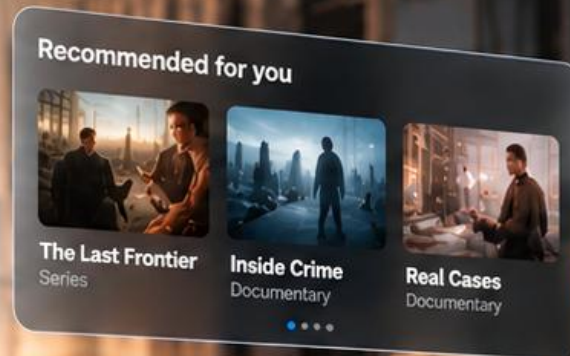
The streaming company uses machine learning to predict Victor's preferences based on data from his activities and those of other users, as well as data collected outside the platform. The system incorporates new data as Victor generates it.

REAL-WORLD EXAMPLE

How data creates value

After a long day at work, Victor loves to watch his favorite shows. Media streaming companies use advanced data analytics to understand his viewing habits and preferences.

Based on these insights, they can provide him with personalized recommendations and create new shows that will appeal to him and other customers like him.

**Analyzes viewing habits**

Duration, genre, watch time, devices

**Compares with similar users**

Patterns from millions of users

**Provides relevant recommendations**

Content that matches his interests

1

**Collects data**

Viewing behavior, searches, and user interactions.



2

**Analyzes patterns**

Uses data analytics and AI to find insights.



3

**Creates value**

Personalized recommendations and new content.



4

**Better experiences**

More relevant shows for Victor and millions of other customers.

Technology and AI

New technologies are changing the way we live, work, and solve problems. From cloud computing to artificial intelligence and connected devices, these innovations help individuals, organizations, and governments create greater value for the people they serve.



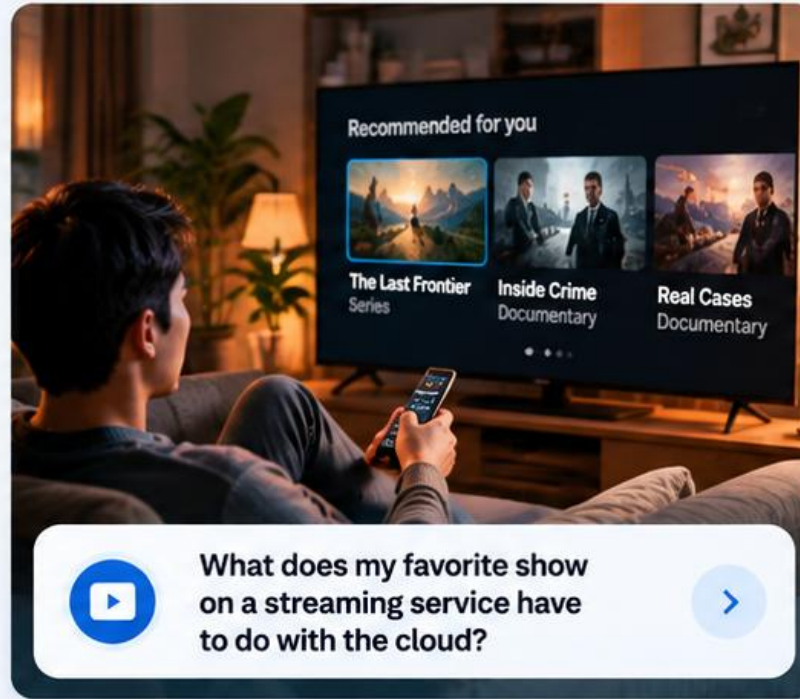
Technology

New technologies—such as cloud, artificial intelligence, and connected devices—help us solve problems at home and at work. The same technologies that improve our personal lives also help companies, non-profits, and governments serve their stakeholders better.



Cloud

The cloud is on-demand computing power and data storage, accessed through the internet. Organizations that use the cloud can adjust their data and processing capacity up or down as needed. Modern applications and business services based in the cloud allow multiple people in different locations to work on the same document simultaneously.



What does my favorite show on a streaming service have to do with the cloud?



REAL-WORLD EXAMPLES

Streaming services use the cloud to deliver content to millions of viewers

The cloud provides on-demand access to data, allowing streaming services to store and deliver millions of shows and movies simultaneously. Customers can find programs to watch and get personalized recommendations based on their interests.



How does the cloud help humanitarian organizations respond to crises?



Humanitarian organizations use cloud-based solutions to respond to crises

Because data is stored in the cloud, volunteers around the world can access the same information and collaborate in real time, helping them quickly find and deliver resources to the people who need them most.



Artificial Intelligence

Artificial intelligence (AI) refers to machines that have been programmed to imitate human intelligence. Examples include speech recognition, image analysis, and predictive analytics that help solve complex problems such as customer retention.



AI in everyday life and business

AI is already part of our daily lives—from virtual assistants like Siri, to personalized recommendations on streaming platforms. Organizations across industries use AI to make better decisions, improve efficiency, and create more personalized experiences for their customers.



Generative AI

Generative AI tools—AI systems capable of creating new and diverse content such as text, images, videos, music, code, and even 3D models—have rapidly advanced and become widely accessible. These tools not only generate creative works but also assist with workflow management, complex problem-solving, personalized learning, and interactive experiences.

Check the additional resources in the Wrap Up lesson at the end of this course if you'd like to delve deeper into the transformative potential of AI.

REAL-WORLD EXAMPLE

How do retailers manage same day delivery using artificial intelligence?

Retailers use AI to predict demand, optimize inventory, and plan delivery routes in real time. By analyzing customer data, traffic conditions, and store inventory, AI helps ensure products can be delivered faster and more efficiently.

See how it works →

AI Optimizing Delivery Routes



REAL-WORLD EXAMPLE

How does generative AI help office workers structure documents in a clear and concise way?

Generative AI can analyze existing data and generate well-structured document drafts, summaries, and key insights. It helps office workers save time, improve clarity, and communicate their ideas more effectively.

Try an example →

Generate a document

Create a concise executive summary based on this data...

Generate

Executive Summary

- Summarize
- Rewrite
- Improve clarity
- Create slide
- Translate

EXAMPLES OF GENERATIVE AI CAPABILITIES



Text

Write, summarize, and brainstorm ideas.

Draft a marketing email for a new product launch...



Image

Create visuals from a simple description.

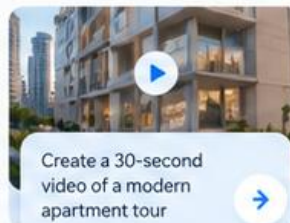


A modern co-living space in a tropical city



Video

Generate videos from text or images.

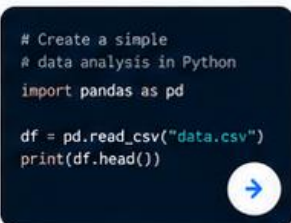


Create a 30-second video of a modern apartment tour



Code

Generate and explain code.



3D Models

Create 3D models from text or images.



A modern chair in minimalist style



Connected Devices

Connected devices are physical objects that use the internet to connect with each other.



Always Connected

Connected devices use the internet to exchange data in real time.



Smarter and More Efficient

With data, connected devices can operate more automatically, accurately, and efficiently.



Creating Real Value

Connected devices help individuals, businesses, and governments deliver safer, more convenient, and more sustainable services.

REAL-WORLD EXAMPLE

How can connected devices help reduce the number of road accidents?

Sensors, cameras, and connected devices on vehicles and road infrastructure can share real-time data, helping to detect potential hazards, provide early warnings, and enable safer, more autonomous driving, ultimately reducing traffic accidents.

[See how it works](#) →



REAL-WORLD EXAMPLE

How do utility companies use connected devices to provide more consistent services?

Connected devices such as smart meters in homes and businesses and sensors across the power grid allow utility companies to monitor usage in real time, detect faults early, reduce outages, and better balance supply and demand.

[See an example](#) →



EXAMPLES OF CONNECTED DEVICES IN EVERYDAY LIFE



Connected Vehicles

Share data with other vehicles and infrastructure to enable safer and more efficient journeys.



Smart Homes

Devices like thermostats, cameras, and smart appliances can be monitored and controlled remotely.



Industry

Sensors on machines monitor performance and help prevent downtime.



Healthcare

Wearable devices monitor vital signs, activity, and health conditions in real time.



Smart Cities

Connected streetlights, public transport, and sensors help create more efficient and livable cities.



New ways of working – Design thinking

The third component of going digital is introducing new ways of working. By using technology and data to collaborate, take smarter risks, and test ideas faster, we can better meet people's needs. Next, we'll explore one powerful way of working: Design Thinking.



What is Design Thinking?

Design thinking is a human-centered way of working focused on understanding the needs of users or customers. It helps us solve problems in a creative and innovative way. It starts with a deep understanding of our users and the problems they face, so we can design products or services that better meet their needs.



Identifying the end user

A key strength of design thinking is its emphasis on truly knowing the end users. We can use various methods—such as interviews, focus groups, observation, and surveys—to learn about their behaviors, needs, and pain points. By understanding what they do, think, and feel, we can create solutions that deliver greater value and a better experience.

REAL-WORLD EXAMPLE

If you are designing hotel rooms, who is your end user?

End users are the guests who stay at the hotel. Understanding their needs, preferences, and behaviors helps us design room features and services that create a more comfortable and memorable stay.



Comfortable and restful stay



Convenient and seamless experience



Personalized services

Explore the example



REAL-WORLD EXAMPLE

If you are testing a new menu item at your restaurant, who is your end user?

End users are your customers. Understanding their tastes, dietary preferences, and dining behaviors helps you design and test menu items that they will enjoy and keep coming back for.



Delicious and high-quality taste



Healthy and nutritious options



Great value and dining experience

See the example



THE DESIGN THINKING PROCESS

01



Empathize

Understand the end users and their needs.



Conduct interviews with hotel guests



02



Define

Identify key problems and opportunities.



Identify common pain points



03



Ideate

Generate many creative ideas.



Brainstorm room concepts



04



Prototype

Build a simple version of the solution.



Create a mock-up room layout



05



Test

Get feedback from users and iterate.



Test with real guests and gather feedback



Solving the right problem

By putting end users at the center, you can uncover unmet needs that should be the focus of your design. No matter your role or industry, understanding how end users feel and defining the right problem saves time and improves the solution.



Start with real people

Go beyond assumptions. Understand what end users do, think, feel, and struggle with.



Find the real problem

Look for unmet needs and root causes, not just surface-level symptoms.



Design better solutions

By solving the right problem, you can create solutions that are more relevant, effective, and impactful.

Government



Rethinking meal delivery for the elderly

A government-run meal delivery service noticed that many elderly recipients were not eating the provided meals.



Initial assumption

The issue was the food menu.



Real problem

The real issue was social isolation. They needed human interaction, not just food.

Food & Beverage



Understanding customer preferences

Milkshake sales were flat at a global fast food chain. The team initially assumed the flavors were the problem.



Initial assumption

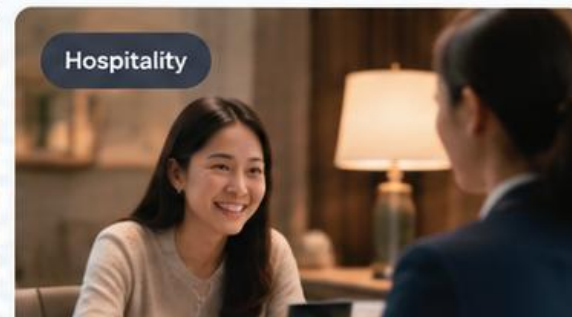
The issue was the texture and flavor options.



Real problem

Customers wanted a more filling, long-lasting, and easy-to-eat option while on the go.

Hospitality



Redesigning the hotel experience

Guests complained about long check-in times. The hotel first thought the issue was the staffing process.



Initial assumption

The issue was staff efficiency.



Real problem

Guests wanted a more seamless and personalized experience, including digital check-in and clearer communication.

KEY TAKEAWAYS

01



Observe

Spend time with end users in their real environment.



02



Define the problem

Identify unmet needs and key pain points.



03



Generate insights

Look for patterns and deeper human needs.



04



Design and test

Create and test solutions with real users.



Applying Design Thinking, for every role

You might be thinking that design thinking is only for people in creative roles. Not true!

No matter what your role or industry is, you can apply the design thinking process to improve your users' experience.



Explore the examples on the right

to see how people from different roles use design thinking in their daily work, even if their roles are not focused on design.

Customer Service Representative



“

When I'm on the phone with customers, I take notes on all the things they wish for. Every month, my boss asks our team to share what we're hearing. She uses these ideas to make our processes better.”

Customer Service Representative

Impact of using design thinking

- ✓ Identifies customer needs directly
- ✓ Improves service processes based on real input
- ✓ Creates a more consistent and better customer experience

Finance Analyst



“

I see the colleagues who read my financial reports as end users. Understanding the information they want and need helps me deliver a more targeted and relevant analysis.”

Finance Analyst

Impact of using design thinking

- ✓ Understands user needs (internal stakeholders)
- ✓ Presents analysis that is more relevant and easier to understand
- ✓ Supports better and faster decision-making

Recruiter



“

I think of job applicants as end users. Personas help me understand who the applicants are and what motivates them. Thinking of our job application process as a user journey helped me create a better experience for our applicants and get them excited to work here.”

Recruiter

Impact of using design thinking

- ✓ Understands candidate needs and motivations
- ✓ Designs a more relevant and efficient recruitment process
- ✓ Creates a more positive candidate experience and stronger talent attraction

New ways of working – Agile

Once you've identified the right problems and have a deeper understanding of your end user, what are the best ways to create and test your ideas?

Agile provides principles and methodologies that help organizations respond rapidly to changing conditions. At the core of Agile are “sprints,” short cycles that speed innovation.

Understand the problem

Test ideas quickly

Deliver real impact

How is Agile different from traditional ways of working?

Traditional way of working

Linear, rigid, and slower

 **Detailed planning upfront**
Plan everything before execution

 **Fixed scope**
Changes are difficult and time-consuming

 **Long development cycles**
Deliver results after a long period

 **Higher risk of misalignment**
May not fully meet user needs

vs.

Agile way of working

Iterative, flexible, and faster

 **Start with user needs**
Focus on solving the right problem

 **Iterative development**
Build, test, and learn in short cycles (sprints)

 **Adapt quickly**
Be responsive to changing conditions

 **Deliver meaningful impact**
Continuous improvement based on real user feedback

How Agile works: The sprint cycle

Sprints are short, time-boxed cycles (typically 1–4 weeks) to turn ideas into real solutions, test with users, and continuously improve.

